

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12419058
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Wang; Xiaoguang et al.

Semiconductor structure, manufacturing method therefor and memory

Abstract

A semiconductor structure, a manufacturing method therefor and a memory are provided. The semiconductor structure may at least include: a plurality of aligned transistors, in which the transistors share a same source plate, channels of the transistors are located above the source plate, the channel length direction of the transistors is perpendicular to a surface of the source plate, and a material of the channels includes a single crystal semiconductor; a plurality of drain contacts, electrically connected with drains of the transistors, in which even number of the transistors share one same drain contact; and a plurality of magnetic tunnel junctions, located on the drain contacts and electrically connected with the drain contacts in one-to-one correspondence.

Inventors: Wang; Xiaoguang (Hefei, CN), Zeng; Dinggui (Hefei, CN), Li; Huihui (Hefei, CN), Cao; Kanyu (Hefei, CN)

Applicant: CHANGXIN MEMORY TECHNOLOGIES, INC. (Hefei, CN); BEIJING SUPERSTRING ACADEMY OF MEMORY TECHNOLOGY (BeiJing, CN)

Family ID: 1000008818432

Assignee: CHANGXIN MEMORY TECHNOLOGIES, INC. (Hefei, CN); BEIJING SUPERSTRING ACADEMY OF MEMORY TECHNOLOGY (Beijing, CN)

Appl. No.: 17/857219

Filed: July 05, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230061246 A1	Mar. 02, 2023

Foreign Application Priority Data

CN	202111020954.5	Sep. 01, 2021
----	----------------	---------------

Related U.S. Application Data

continuation parent-doc WO PCT/CN2022/077730 20220224 PENDING child-doc US 17857219

Publication Classification

Int. Cl.: H10B61/00 (20230101); H10N50/01 (20230101)

U.S. Cl.:

CPC H10B61/22 (20230201); H10N50/01 (20230201);

Field of Classification Search

CPC: H10B (61/22); H10N (50/01); H10N (50/10); H01L (27/228); H01L (43/02); H01L (43/08); H01L (43/12); H01L (29/42392); H01L (29/7869); H01L (29/78696)

USPC: 257/541

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7781285	12/2009	Kim	N/A	N/A
8542519	12/2012	Asao	N/A	N/A
8830735	12/2013	Kariyada	N/A	N/A
8878156	12/2013	Satoh	N/A	N/A
9029822	12/2014	Satoh	N/A	N/A
9117749	12/2014	Or-Bach	N/A	N/A
10224279	12/2018	Or-Bach	N/A	N/A
10297290	12/2018	Beigel	N/A	N/A
10446606	12/2018	Kumar	N/A	N/A
10672432	12/2019	Beigel	N/A	N/A
11127783	12/2020	Zhu	N/A	N/A
2007/0080385	12/2006	Kim	N/A	N/A
2010/0059837	12/2009	Kim	N/A	N/A
2010/0289084	12/2009	Yoon	N/A	N/A
2011/0215382	12/2010	Asao	N/A	N/A
2011/0266600	12/2010	Kanaya	257/295	H01L 21/76897
2012/0135275	12/2011	Kariyada	N/A	N/A
2012/0306004	12/2011	Yoon	N/A	N/A
2014/0138609	12/2013	Satoh	257/5	H10B 61/22
2014/0346518	12/2013	Kariyada et al.	N/A	N/A
2015/0340316	12/2014	Or-Bach et al.	N/A	N/A
2016/0043136	12/2015	Kim et al.	N/A	N/A
2019/0027535	12/2018	Kumar et al.	N/A	N/A
2019/0157345	12/2018	Zhu et al.	N/A	N/A
2019/0206449	12/2018	Beigel et al.	N/A	N/A
2019/0259807	12/2018	Kumar et al.	N/A	N/A

2019/0348464	12/2018	Sukekawa	N/A	H01F 10/329
2020/0144330	12/2019	Majhi	N/A	H10D 30/6755
2020/0286528	12/2019	Beigel et al.	N/A	N/A
2021/0313391	12/2020	Kumar et al.	N/A	N/A
2021/0408118	12/2020	Zhu et al.	N/A	N/A
2022/0013151	12/2021	Beigel et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102479541	12/2011	CN	N/A
109461756	12/2018	CN	N/A
110571261	12/2018	CN	N/A
110892531	12/2019	CN	N/A
111816673	12/2019	CN	N/A
212136451	12/2019	CN	N/A
212391999	12/2020	CN	N/A
112349322	12/2020	CN	N/A
3958319	12/2021	EP	N/A
2017052542	12/2016	WO	N/A
2018182724	12/2017	WO	N/A

Primary Examiner: Erdem; Fazli

Attorney, Agent or Firm: SYNCODA LLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The application is a continuation application of International Application No.: PCT/CN2022/077730, filed on Feb. 24, 2022, which claims priority to Chinese Patent Application No. 202111020954.5, filed on Sep. 1, 2021. The disclosures of International Application No.: PCT/CN2022/077730 and Chinese Patent Application No. 202111020954.5 are hereby incorporated by reference in their entireties.

BACKGROUND

(1) Semiconductor memory is used in most electronic devices to store data. Semiconductor memory includes dynamic random access memory (DRAM) and magnetic random access memory (MRAM). The data storage operation needs a certain driving current. However, with the miniaturization of semiconductor devices, the interconnection resistance of adjacent conductors increases with the decrease of contact area, and the driving current decreases under the same driving voltage, which leads to the inability to maintain the original driving ability. In addition, when the driving current is maintained at the original level by increasing the voltage supplied by the voltage source, the power consumption of the semiconductor device increases.

SUMMARY

(2) Embodiments of the disclosure relate to, but are not limited to, a semiconductor structure, a method for manufacturing the semiconductor structure and a memory.

(3) In one aspect, embodiments of the disclosure provide a semiconductor structure, which at least includes: a plurality of aligned transistors, in which the transistors share a same source plate, channels of the transistors are located above the source plate, a channel length direction of the transistors is perpendicular to a surface of the source plate, and a material of the channels includes a single crystal semiconductor; a plurality of drain contacts, electrically connected with drains of

the transistors, in which even number of the transistors share one same drain contact; and a plurality of magnetic tunnel junctions, located on the drain contacts and electrically connected with the drain contacts in one-to-one correspondence.

(4) According to some embodiments of the disclosure, another aspect of the embodiments of the disclosure also provides a memory, which at least includes the semiconductor structure described in any one of the above.

(5) According to some embodiments of the disclosure, a further aspect of the embodiments of that disclosure also provides a manufacturing method of a semiconductor structure, including the following operations. A plurality of aligned transistors are provided, in which the transistors share a same source plate, channels of the transistors are located above the source plate, a channel length direction of the transistors is perpendicular to a surface of the source plate, and a material of the channels includes a single crystal semiconductor. A plurality of drain contacts are formed, and electrically connected with drain of the transistor, in which even number of the transistors share one same drain contact. A plurality of magnetic tunnel junctions are formed, which are located on the drain contacts, and electrically connected with the drain contacts in one-to-one correspondence.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) One or more embodiments are illustrated by the corresponding pictures in the drawings, which do not constitute a limitation on the embodiments. Unless otherwise stated, the figures in the drawings do not constitute a limitation of scale.

(2) FIGS. **1** to **6** are structural scheme diagrams of a semiconductor structure provided by an embodiment of the disclosure.

(3) FIGS. **7** to **12** are structural scheme diagrams corresponding to each operation of a manufacturing method of a semiconductor structure provided by an embodiment of the disclosure.

DETAILED DESCRIPTION

(4) Embodiments of the disclosure will be described in detail below with reference to the accompanying drawings. However, one of ordinary skill in the art will understand that, numerous technical details have been set forth in various embodiments of the disclosure, in order to make the reader better understand the disclosure. However, even without these technical details and various changes and modifications based on the following embodiments, the technical solutions claimed in this disclosure can be realized.

(5) FIGS. **1** to **6** are structural scheme diagrams of a semiconductor structure provided by an embodiment of the disclosure.

(6) Referring to FIGS. **1** to **3**, the semiconductor structure includes a plurality of aligned transistors, which share a same source plate **12**, channels **13** of the transistors are located on the source plate **12**, a direction along channel length of the transistors is perpendicular to a surface of the source plate **12**, and a material of the channels **13** includes a single crystal semiconductor; a plurality of drain contacts **20** connected to drains **19** of the transistors, in which an even number of the transistors share one same drain contact **20**; and a plurality of magnetic tunnel junctions **22**, located on the drain contacts **20** and electrically connected to the drain contacts **20** in one-to-one correspondence.

(7) The embodiment of the disclosure will be described in more details below with reference to the accompanying drawings.

(8) The semiconductor structure includes a substrate **11** and a source plate **12** located on the substrate **11**. The substrate **11** may be composed of an insulating material to function for isolation, and may also be a silicon wafer to facilitate growth of a single crystal material. The source plate **12** may be independent of the transistors, act as a source contact and provide source signals to the

transistors, or act as sources of the transistors to receive source signals. In any case, a material of the source plate **12** is a conductive material, such as at least one of doped silicon, indium tin oxide (ITO), metal molybdenum, metal aluminum, metal titanium or metal copper.

(9) In some embodiments, the channels **13** of the transistors are in direct contact with the source plate **12**, and the channels **13** are integrated with the source plate **12**, that is to say, the source plate **12** serves as sources of the transistors, and the plurality of the aligned transistors share the same source plate **12** and the same source signal, so that complexity of the process is reduced without using a mask and an etching process to prepare a plurality of discrete sources. The source plate **12** and the channels **13** can include the same single crystal semiconductor, so that the contact resistances between the source plate **12** and the channels **13** can be reduced, and each transistor can be guaranteed to have a larger driving current under a preset driving voltage. Meanwhile, the single crystal semiconductor includes single crystal silicon or single crystal germanium, and the source plate **12** includes doped single crystal silicon or doped single crystal germanium.

(10) In some embodiments, the semiconductor structure includes a substrate **11**, a source plate **12**, a first isolation layer **15**, word lines **16**, and a second isolation layer **17** that are stacked in sequence. The first isolation layer **15** is disposed between the word lines **16** and the source plate **12** for isolating the source plate **12** from the word lines **16**. The second isolation layer **17** is disposed between adjacent word lines **16** and covers top surfaces of the word lines **16**, mainly for isolating word lines **16** from each other, as well as isolating the word lines **16** from other stacked film layers above the second isolation layer **17**. Said other stacked film layers include but are not limited to the drain contacts **20**.

(11) It should be understood that a material of the first isolation layer **15** and a material of the second isolation layer **17** may be same or different. Meanwhile, the second isolation layer **17** may be composed of a first portion and a second portion. The first portion is disposed between adjacent word lines **16**, and the second portion is disposed between the word lines **16** and the other stacked film layers. A material of the first portion and a material of the second portion may be same or different. For example, a dielectric constant of the material of the first portion is smaller than a dielectric constant of the material of the second portion. The first portion is mainly configured to suppress signal crosstalk of the adjacent word lines **16**, and the second portion is configured to suppress signal crosstalk of the adjacent transistors and short circuit of film layers on opposite sides.

(12) In addition, gate dielectric layers **14** and channels **13** pass through the second isolation layer **17**, the word lines **16** and the first isolation layer **15** sequentially. Bottoms of the gate dielectric layers **14** and the channels **13** are in direct contact with the surface of the source plate **12**. Each gate dielectric layer **14** covers entire sidewall surface of each channel **13**. A length of the gate dielectric layers **14** and the channels **13** in the direction perpendicular to the surface of the source plate **12** is equal to the sum of the thicknesses of the first isolation layer **15**, the word lines **16**, and the second isolation layer **17**. The length of the gate dielectric layers **14** is equal to the length of the channels **13**. In other embodiments, a gate dielectric layer **14** is located only between a channel **13** and a word line **16**, the gate dielectric layer **14** covers only part of the sidewall of the channel **13**, and the length of the gate dielectric layer **14** is equal to the thickness of the word line **16** in a direction perpendicular to the surface of the source plate **12**.

(13) In some embodiments, a material of the gate dielectric layer **14** is a material with a high dielectric constant. For example, the material of the gate dielectric layer **14** includes at least one of HfO_2 , ZrO_2 or HfON . The material of the gate dielectric layer **14** is provided with a high dielectric constant, which is conducive to suppressing the gate leakage current caused by electron tunneling in the case that the thickness of the gate dielectric layers **14** is thin.

(14) Referring to FIG. 4, the semiconductor structure includes a plurality of word lines **16** arranged in sequence, extending in a first direction **D1** and arranged in a second direction **D2** perpendicular to the first direction **D1**. Each word line **16** is configured to drive the transistors in its extending

direction, and surrounds the channels **13**. It can be understood that since the word line **16** surround the channels **13**, a width d of each word line **16** is greater than an outer diameter of each gate dielectric layer **14** in the second direction **D2**.

(15) In some embodiments, the transistors are arranged in an aligned array. The array of the transistors may be considered to be consisted of different rows extending in the first direction **D1** and arranged in the second direction **D2**. Transistors of a row are aligned in the first direction **D1**, and adjacent rows are equally spaced in the second direction **D2**. The distances of adjacent transistors in the same row may be equal to the distances of adjacent rows in the second direction **D2**. Each word line **16** is configured to drive all the transistors in a same row in the array of the transistors.

(16) In some embodiments, multiple transistors sharing one same drain contact **20** are located in at least two rows of the array of the transistors, and accordingly, the multiple transistors connected to the same drain contact **20** are driven by at least two word lines **16**. Each of the drain contacts **20** has a regular pattern such as rectangle or square. A plurality of the drain contacts **20** are arranged in a regular array. The geometrical centers of the plurality of the drain contacts **20** in a same row are in a same straight line, so that when the center of the projection of a magnetic tunnel junction and the center of the projection of the corresponding drain contact coincide, the bit lines **23** can be effectively and uniformly contacted with each magnetic tunnel junction **22** in the extending direction thereof, thereby ensuring the signal transmission stability of the semiconductor structure. In other words, when the geometric centers of the plurality of the drain contacts **20** in the same row are in the same line, under the condition that the position of the bit line **23** is not changed and effective contact between the bit line **23** and the magnetic tunnel junctions **22** is ensured, the projection of the center of a magnetic tunnel junction **22** can coincide with the projection of the center of the corresponding drain contact **20**, thereby ensuring that the drain contact **20** and the magnetic tunnel junction **22** have a good signal transmission performance.

(17) For example, referring to FIG. 5, 2M transistors share a same drain contact **20**. The 2M transistors are located in two adjacent rows of the array of the transistors, each row has corresponding M transistors, and each drain contact **20** is rectangular, in which M is an integer greater than or equal to 2. For example, when M is equal to 3, 6 transistors share one same drain contact **20**. Referring to FIG. 6, $N \times 2$ transistors share one same drain contact **20**, the drain contact **20** is square, in which N is an integer greater than or equal to 2. For example, when N is equal to 2, four transistors share the same drain contact **20**.

(18) It should be understood that multiple transistors sharing a same drain contact **20** may also be located in a same row of the array of the transistors. In addition, based on the above description, KM transistors may share a same drain contact **20**, and both K and M are parameters greater than or equal to 2. When K equals 2, as shown in FIG. 5, multiple transistors sharing the same drain contact **20** are located in two rows of the array of the transistors; when K equals 3, multiple transistors sharing the same drain contact **20** are located in three rows of the array of the transistors; and when K equals M, referring to FIG. 6, the drain contact **20** has a square shape.

(19) In some embodiments, the drain **19** of each transistor is located between the channel **13** and the drain contact **20**, and each transistor has a corresponding drain **19**, that is, the drains **19** corresponding to different transistors are relatively independent. In other embodiments, the drains **19** have the same shape as the drain contacts **20**, that is, even-number of the transistors share a same drain. In the direction perpendicular to the surface of the source plate **12**, an orthographic projection of each drain contact **20** coincides with an orthographic projection of each drain **19**. By setting the shape of each drain **19** to be the same as the shape of each drain contact **20**, it is advantageous to increase the contact area between the drain **19** and the drain contact **20** and reduce the contact resistance between the drain **19** and the drain contact **20**, thereby ensuring that the magnetic tunnel junction **22** can receive a larger driving current under a preset driving voltage.

(20) In some embodiments, in the direction perpendicular to the surface of the source plate **12**, each

magnetic tunnel junction **22** includes a fixed layer **221**, a tunnel junction **222**, and a free layer **223** stacked in sequence. The fix layer **221** covers a surface of one drain contact **20**. The fixed layer **221** and the free layer **223** are formed of a ferromagnetic material having in-plane magnetic anisotropy or having perpendicular magnetic anisotropy, and a magnetic orientation of the fixed layer **221** remains fixed. When magnetic orientations of the free layer **223** and the fixed layer **221** are adversely parallel (AP) to each other, a first state (for example, logic “1”) is stored, and when the magnetic orientations of the free layer **223** and the fixed layer **221** are parallel (P) to each other, a second state (for example, logic “0”) is stored. The relationship of the magnetic orientations of the free layer **223** and the fixed layer **221** can be judged by sensing a resistance when a current flows through the magnetic tunnel junction **22**, thereby implementing data reading. Accordingly a transistor can be turned on by activating the corresponding word line **16**, so as to generate a write current (i.e. the drive current described above) flowing through the magnetic tunnel junction **22** which adjusts the magnetic orientation of the free layer **223** to implement data storage.

(21) In some embodiments, the semiconductor structure further includes a plurality of bit lines **23** arranged in sequence. The bit lines **23** are arranged in a direction perpendicular to the arrangement direction of the word lines **16**. Each of the bit lines **23** connects all the magnetic tunnel junctions **22** in the extending direction of the bit line. The bit lines **23** are above the magnetic tunnel junctions **22**. By arranging the bit lines **23** above the magnetic tunnel junctions **22**, it is advantageous to avoid that the cross-sectional size of the bit lines **23** is limited by other structures on the same plane, and to ensure that the bit lines **23** have a wider size along its arrangement direction and a larger cross-sectional area along its extending direction, thereby ensuring that the bit lines **23** have lower transmission resistances and better signal transmission performances.

(22) In the embodiment of the disclosure, the semiconductor structure adopts a single crystal semiconductor as the material of the channels of the transistors, as such, since the single crystal semiconductor has high carrier mobility, the transistor device prepared based on the single crystal semiconductor has high switching performance. In addition, when each drain contact is connected to an even number of transistors at the same time, the shape of the drain contact may be a regular rectangle, and the centers of a plurality of drain contacts in a same row is in the same straight line, so that the center projection of each magnetic tunnel junction and the center projection of the corresponding drain contact are coincided. Each bit line is effectively and uniformly contacted with each magnetic tunnel junction in the extending direction of the bit line, thereby ensuring the stable signal transmission performance of the semiconductor structure.

(23) The embodiments of the disclosure also provide a memory, including any one of the above semiconductor structures. The memory prepared based on the above-mentioned semiconductor structure has lower requirements on driving voltage and less self-loss, which is conducive to reducing the heat generation during the operation of an electronic device, thereby reducing the cooling cost of the electronic device and improving the competitiveness of the electronic equipment.

(24) FIGS. **7** to **12** show structural scheme diagrams corresponding to each operation of a manufacturing method of a semiconductor structure provided by an embodiment of the disclosure. Referring to FIGS. **7** to **12**, the manufacturing method of the semiconductor structure provided by the embodiment of the disclosure includes the following operations.

(25) Referring to FIG. **7**, a substrate **11**, a source plate **12** and an intrinsic layer **13a** are provided stacked sequentially.

(26) The substrate **11** may be either a supporting material for supporting formation of subsequent films, or a collection of all the formed films, only schematically shown as the substrate. When the substrate **11** only plays a supporting role, the substrate **11** may be removed after the target structure is formed, or the substrate **11** may not be formed. The substrate **11** may be a silicon wafer to facilitate growth of a single crystal material. The source plate **12** may serve as a source of transistors subsequently formed, or provide a source signal to the transistors subsequently formed.

A material of the source plate **12** is a conductive material, such as at least one of doped silicon (single crystal silicon, polysilicon or microcrystalline silicon), indium tin oxide (ITO), metal molybdenum, metal aluminum, metal titanium or metal copper. A material of the intrinsic layer **13a** includes a single crystal semiconductor and the intrinsic layer **13a** is configured to prepare channels of the transistors.

(27) Referring to FIG. **8**, the intrinsic layer **13a** (refer to FIG. **7**) is etched to form a plurality of aligned channels **13**.

(28) In some embodiments, the channels **13** of the transistors are in direct contact with the source plate **12**, which serves as the sources of the transistors, and the plurality of aligned transistors share the same source plate **12** and share the same source signal. The source plate **12** and the channels **13** may include the same single crystal semiconductor to reduce contact resistance between the source plate **12** and the channels **13**.

(29) In some embodiments, the single crystal semiconductor includes single crystal germanium or single crystal silicon. It is conducive to improving the carrier transfer rate of a transistor device by using single crystal silicon or single crystal germanium as the material of the intrinsic layer **13a**, and improving switching performance of the array of the transistors. In addition, the channels are formed first, and then an isolation layer located between adjacent transistors is formed, which is conducive to enlarging the diffusion space of deposited materials, so that a deposited gate dielectric layer can be more uniformly attached to the sidewall of each channel, so as to ensure that thicknesses of the gate dielectric layers in different transistors and thicknesses of different positions of a same transistor are similar, and ensure that the transistors have stable and reliable electrical properties.

(30) Referring to FIG. **9**, a gate dielectric layer **14** is formed around the sidewall of each channel **13**.

(31) In some embodiments, a material of the gate dielectric layer **14** is a material with a high dielectric constant. For example, the material of the gate dielectric layer **14** includes HfO_2 , ZrO_2 or HfON . It is conducive to suppressing gate leakage current caused by electron tunneling by using the material with a high dielectric constant as the gate dielectric layers **14**, when the thickness of the gate dielectric layers **14** is thin.

(32) Referring to FIG. **10**, a first isolation layer **15**, a plurality of word lines **16** arranged in sequence and a second isolation layer **17** are formed.

(33) A first isolation layer **15** located between the word lines **16** and the source plate **12** is configured to isolate the source plate **12** and the word lines **16**, and to suppress signal crosstalk of adjacent transistors. The word lines **16** extend in the first direction **D1** and are arranged along a second direction **D2**. Each word line **16** is configured to connect with and drive all the transistors located along the extending direction, in which the word line **16** surrounds the channels of the transistors to be driven, and the width of the word line **16** in the second direction **D2** is greater than the outer diameter of the gate dielectric layers **14**. A second isolation layer **17** is filled between the adjacent word lines **16** and covers on the word lines, top surfaces of the channels **13** are flush with a top surface of the second isolation layer **17**, and part of an isolation material filled between the adjacent word lines **16** and part of an isolation material covering the top surfaces of the word lines **16** may be different.

(34) Materials of the first isolation layer **15** and the second isolation layer **17** are both materials with low dielectric constants, including but not limited to silicon oxide, silicon nitride, silicon carbide or silicon oxynitride.

(35) Referring to FIG. **11**, a third isolation layer **18** and drains **19** are formed, in which a drain **19**, a channel **13**, the source plate **12**, a gate dielectric layer **14** and part of a word line **16** constitute a transistor.

(36) In some embodiments, even number of the channels **13** share a same drain **19**, and the multiple transistors sharing the same drain **19** also share a same drain contact **20**. In a direction

perpendicular to the surface of the source plate **12**, the drains **19** may have the same shape as the drain contacts **20** formed subsequently. In addition, the drains **19** and the channels **13** may contain the same single crystal semiconductor. For example, the channels **13** are of single crystal silicon and the drains **19** are of doped single crystal silicon. In other embodiments, each transistor corresponds to a drain **19**, and the drains **19** corresponding to different transistors are independent of each other. Moreover, the material of the drains **19** may be the same as that of the source plate **12**.

(37) In some embodiments, the drains **19** and the channels **13** may contain the same single crystal semiconductor, and a third isolation layer **18** is formed before the drains **19** are formed. It can be understood that, since the drains **19** and the channels **13** contain the same material, an etchant for the drains **19** has a faster etching rate for the channels **13**. In the case that a conductive layer is formed first and then etched to form the drains **19**, the etching process may cause damage to the channels **13**, especially when the drains **19** do not completely cover the top surfaces of the channels **13**. Furthermore, the material of the third isolation layer **18** may be the same as the material of the first isolation layer **15** or the material of the second isolation layer **17**.

(38) Referring to FIG. **12**, drain contacts **20** and a fourth isolation layer **21** are formed.

(39) In some embodiments, the drain contacts **20** have the same shape as the connected drains. Since contact resistance between the drains **19** and subsequently formed magnetic tunnel junctions **22** is larger than contact resistance between the drain contacts **20** and the magnetic tunnel junctions **22**, it is conducive to reducing the contact resistances between different conductors by providing the drain contacts **20**, thereby increasing driving current of the magnetic tunnel junctions **22** under a fixed driving voltage. In the case that the drains **19** of different transistors are relatively independent, providing the drain contacts **20** is also conducive to increasing contact area between the drains and the magnetic tunnel junctions, and supporting the magnetic tunnel junctions, so as to ensure that the magnetic tunnel junctions **22** of any size can be effectively connected with the corresponding transistors.

(40) In some embodiments, the third isolation layer **18** and the fourth isolation layer **21** are two separate film layers, and a material of the fourth isolation layer **21** is the same as a material of the third isolation layer **18**. In another embodiment, the drain contacts **20** are formed after the drains **19** are formed, and the isolation layers **18**, **21** are formed after the drain contacts **20** are formed. The isolation layers **18**, **21** are filled between the adjacent drains **19** and between the adjacent drain contacts, respectively, that is, the isolation layer **18** filled between the adjacent drains and the isolation layer **21** filled between the adjacent drain contacts are integrated.

(41) Referring to FIGS. **1** to **3**, the magnetic tunnel junctions **22** and bit lines **23** are formed.

(42) The magnetic tunnel junctions **22** are located on the drain contacts **20**, and the magnetic tunnel junctions **22** are electrically connected to the drain contacts **20** in one-to-one correspondence. The bit lines **23** are located above the magnetic tunnel junctions **22**. The extending direction of the bit lines **23** is perpendicular to the extending direction of the word lines **16**, and the arrangement direction of the bit lines **23** is perpendicular to the arrangement direction of the word lines **16**. Each bit line **23** connects all the magnetic tunnel junctions **22** along its extending direction.

(43) In the embodiment of the disclosure, the semiconductor structure adopts a single crystal semiconductor as the material of the channels of the transistors. Because the single crystal semiconductor has higher carrier mobility, a transistor device prepared based on the single crystal semiconductor has higher switching performance. In addition, when a drain contact is connected to even number of the transistors at the same time, the shape of the drain contact may be a regular rectangle, and the centers of a plurality of the drain contacts in a same row is in a same straight line, so that the center projection of each magnetic tunnel junction and the center projection of each corresponding drain contact are coincided, each bit line is effectively and uniformly contacted with each magnetic tunnel junctions in the extending direction of the bit line, thereby ensuring the stable signal transmission performance of the semiconductor structure.

(44) Those of ordinary skill in the art will understand that, the above-described embodiments are intended to achieve specific embodiments of the disclosure, while in practical application, various changes in forms and details may be made thereto without departing from the spirit and scope of the disclosure. Any person skilled in the art may make their own changes and modifications without departing from the spirit and scope of the disclosure, so the scope of protection of the disclosure shall be subject to the scope defined by the claims.

INDUSTRIAL PRACTICALITY

(45) In the embodiments of the disclosure, the semiconductor structure may at least include: a plurality of aligned transistors, in which the transistors share a same source plate, channels of the transistors are located above the source plate, a channel length direction of the transistors is perpendicular to a surface of the source plate, and a material of the channels includes a single crystal semiconductor; a plurality of drain contacts, electrically connected with the drains of the transistors, in which even number of the transistors share one same drain contact; and a plurality of magnetic tunnel junctions, located on the drain contacts and electrically connected with the drain contacts in one-to-one correspondence. The embodiments of the disclosure are conducive to improving the electrical properties of semiconductor structures.

Claims

1. A semiconductor structure, comprising: a plurality of aligned transistors, wherein the transistors share a same source plate, channels of the transistors are located above the source plate, a channel length direction of the transistors is perpendicular to a surface of the source plate, and a material of the channels comprises a single crystal semiconductor; a plurality of drain contacts with a regular rectangle, electrically connected with drains of the transistors, wherein even number of the transistors share one same drain contact; and a plurality of magnetic tunnel junctions, located on the drain contacts and electrically connected with the drain contacts in one-to-one correspondence; and a center projection of each magnetic tunnel junction coincides with a center projection of each corresponding drain contact onto the surface of the source plate.
2. The semiconductor structure according to claim 1, wherein the channels of the transistors are in direct contact with the source plate, and the channels are integrated with the source plate.
3. The semiconductor structure according to claim 1, wherein the source plate comprises the single crystal semiconductor.
4. The semiconductor structure according to claim 3, wherein the single crystal semiconductor comprises single crystal silicon or single crystal germanium.
5. The semiconductor structure according to claim 1, further comprising: a plurality of word lines arranged in sequence above the source plate, wherein each of the word lines is configured to drive the transistors in an extending direction of the word line, and surrounds the channels.
6. The semiconductor structure according to claim 5, wherein the plurality of the transistors connected to the same drain contact are driven by at least two of the word lines.
7. The semiconductor structure according to claim 5, further comprising: a plurality of bit lines arranged in sequence, wherein an arrangement direction of the bit lines is perpendicular to an arrangement direction of the word lines, and each of the bit lines connects all of the magnetic tunnel junctions in an extending direction of the bit line, and is located on the magnetic tunnel junctions.
8. The semiconductor structure according to claim 5, further comprising: gate dielectric layers, which are respectively located between channels and word lines and respectively surround the channels, wherein a material of the gate dielectric layers comprises at least one of HfO_2 , ZrO_2 or HfON .
9. The semiconductor structure according to claim 8, wherein in the direction perpendicular to the surface of the source plate, a length of the gate dielectric layers is equal to a length of the channels,

or the length of the gate dielectric layers is equal to a thickness of the word lines.

10. The semiconductor structure according to claim 1, wherein the drain contacts are square or rectangular.

11. The semiconductor structure according to claim 10, wherein N^2 transistors share one same drain contact, and the drain contact is square, wherein N is an integer greater than or equal to 2.

12. The semiconductor structure of claim 10, wherein $2M$ transistors share one same drain contact, and the drain contact is rectangular, wherein M is an integer greater than or equal to 2.

13. A memory, comprising the semiconductor structure according to claim 1.

14. A manufacturing method of a semiconductor structure, comprising: providing a plurality of aligned transistors, wherein the transistors share a same source plate, channels of the transistors are located above the source plate, a channel length direction of the transistors is perpendicular to a surface of the source plate, and a material of the channels includes a single crystal semiconductor; forming a plurality of drain contacts with a regular rectangle electrically connected with drains of the transistors, wherein even number of the transistors share one same drain contact; and forming a plurality of magnetic tunnel junctions located on the drain contacts, and are electrically connected with the drain contacts in one-to-one correspondence; and a center projection of each magnetic tunnel junction coincides with a center projection of each corresponding drain contact onto the surface of the source plate.

15. The manufacturing method of a semiconductor structure according to claim 14, wherein forming the transistors comprises: forming the source plate and an intrinsic layer which are stacked; etching the intrinsic layer to form a plurality of the channels that are aligned arranged; forming gate dielectric layers respectively surrounding the sidewalls of the channels; and forming a first isolation layer and a plurality of word lines arranged in sequence, wherein the first isolation layer is located between the word lines and the source plate, and the word lines respectively surround the gate dielectric layers in an extending direction of the word line.
